

As you have noticed the first 3TN set of lecture notes is on the revision on the real numbers with the concepts of supremum, infimum, minimum, maximum, sequences (in particular recursive sequences). These concepts are the foundation of Mathematical Analysis and therefore need to be properly studied. Our first tutorial is on those concepts.

1. For each of the following subsets of $\mathbb{R}$ find the supremum and the infimum specifying whether these are maximum and minimum respectively.

(a) $A := \left\{ \frac{1+2n}{n} : n \in \mathbb{N} \setminus \{0\} \right\};$

(b) $B := \left\{ (-1)^n \frac{n+1}{n} : n \in \mathbb{N} \setminus \{0\} \right\};$

(c) $C := \left\{ \frac{m}{n} + \frac{n}{m} : m, n \in \mathbb{N} \setminus \{0\} \right\};$

(d) $D := \left\{ \frac{m}{n} - \frac{n}{m} : m, n \in \mathbb{N} \setminus \{0\} \right\};$

(e) $E := \left\{ \frac{x}{x+1} : x \in (2, 3) \right\};$

(f) $F := \left\{ \frac{x}{x-1} : x > 1 \right\}.$

2. Find the following limits:

(a) $\lim_{n \rightarrow \infty} \frac{2}{\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}};$

(b) $\lim_{n \rightarrow \infty} [n \ln(n-1) - (n-1) \ln n].$

3. Prove that if $a_n \rightarrow \ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$. Is the converse true? Explain your answer.

4. Prove that if $a_n \rightarrow \ell \neq 0$ with $\ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$. Is the converse true? Explain your answer.

5. Show that if $a_1, \dots, a_n$ are n real numbers such that $0 \leq a_i < 1 \forall i$, then

$$\prod_{i=1}^n (1 - x_i) \geq 1 - \sum_{i=1}^n x_i.$$

6. Show that the sequence (u_n) defined by $u_1 = 0$ and $u_{n+1} = \sqrt{2 + u_n}$ for any $n = 1, 2, \dots$ converges to 2.

7. Study the behaviour (monotonicity, boundedness, convergence) of the sequence of real numbers defined by $a_0 = 1$, and $a_{n+1} = 1 + \frac{1}{a_n}$ for any n .
(Hint: Consider the subsequences (a_{2n}) and (a_{2n+1}) .)

8. Let (x_n) be a sequence of real numbers defined by
$$\begin{cases} x_0 \in (0, 1], \\ x_{n+1} = \frac{x_n}{2} + \frac{x_n^2}{4} \end{cases} \quad \forall n \in \mathbb{N}$$
- (a) Show that $x_n > 0 \quad \forall n \in \mathbb{N}$.
 (b) Show that $x_n \leq 1 \quad \forall n \in \mathbb{N}$.
 (c) Show that the sequence (x_n) is monotone (increasing/decreasing)
 (d) Deduce that the sequence (x_n) converges and find its limit.

9. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\exists \lim_{x \rightarrow \infty} f(x) = \ell \in \mathbb{R} \quad \Rightarrow \quad \lim_{x \rightarrow \infty} \int_x^{x+1} f(t) dt = \ell.$$

(Hint: You can use the definition of limit at infinity with $\varepsilon > 0$.)

10. Show that $2|xy| \leq \varepsilon x^2 + \frac{1}{\varepsilon} y^2 \quad \forall x, y \in \mathbb{R}, \quad \forall \varepsilon > 0$.
(Hint: Send $2|xy|$ to the right-hand side and identify the identity.)

11. Show that

$$(i) \quad |\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} \quad \forall x, y \in \mathbb{R} \quad (ii) \quad |\sqrt{1 + x^2} - \sqrt{1 + y^2}| \leq |x - y| \quad \forall x, y \in \mathbb{R}.$$